

A ResNet-18 Baseline Study of Distortion-Aware Training for Robust AI-Generated Image Detection

Junxuan Cai (Student ID: 521370910086)
ECE4880J Computer Vision
Global College, Shanghai Jiao Tong University
Shanghai, China

Abstract—Routine post-processing is a practical source of distribution shift for AI-generated image detectors. We measure this shift with two ImageNet-pretrained ResNet-18 training conditions evaluated across three seeds. The baseline and distortion-aware conditions use the same 27,643 labeled images from NTIRE 2026 training shard 5, source-isolated split, optimization schedule, and evaluation sets. Distortion-aware training adds random JPEG compression, Gaussian blur, and Gaussian noise during training. Performance is measured on clean images and five post-processing transformations, using three training seeds, five severity levels per transformation, and single-component training comparisons. The baseline obtains 97.30% clean ROC AUC and 94.95% pooled transformed AUC. Distortion-aware training changes these values to 96.51% and 95.90%, respectively: a 0.79 *pp* clean-test reduction, a 0.95 *pp* transformed-test gain, and a 1.74 *pp* reduction in the clean-to-transformed gap. The direction of this exchange is consistent across seeds. The gain is concentrated under severe Gaussian noise and blur; JPEG compression and center crop show no consistent improvement.

Index Terms—AI-generated image detection, distribution shift, data augmentation, robustness, image forensics, common corruptions

I. Introduction

AI-generated images are commonly recompressed, resized, cropped, or otherwise edited before they reach a detector. These operations may leave the depicted content recognizable while weakening low-level traces used for image forensics. A detector that performs well on original files can therefore lose accuracy after an image passes through a social platform, messaging service, or editing tool. Compression, resizing, and blur have consequently become standard stress tests for synthetic-image detectors [1]–[3].

Post-processing robustness is distinct from cross-generator generalization. The latter asks whether a detector transfers to generators absent from its training set; the former asks whether a prediction survives changes to the same source image. This report studies post-processing. The available shard provides binary real/AI labels but no generator identities, so it does not support a generator-held-out split. The evaluation therefore covers source-separated post-processing shifts [4], [5].

The baseline is an ImageNet-pretrained ResNet-18 trained with crop, flip, and color augmentation. A second

ResNet-18 uses the same images, split, initialization, loss, optimizer, schedule, model-selection rule, and test sets, but adds random JPEG compression, Gaussian blur, and Gaussian noise during training. This isolates the effect of the added distortions without changing model capacity or training budget. Figure 1 gives an overview of the comparison.

The controlled comparison reveals a trade-off: distortion-aware training improves pooled transformed AUC and narrows the clean-to-transformed gap while reducing clean AUC. Transformation-specific and severity results locate the gain mainly in strong noise and blur; JPEG compression and center crop do not improve consistently.

II. Related Work

A. Cross-Generator Detection

Synthetic-image detection has often been studied as transfer from a known generator to unseen generators. Wang et al. trained a binary classifier on ProGAN samples and found that blur and JPEG augmentation helped it transfer to several CNN-based generators, although performance remained generator-dependent [1]. Diffusion models introduced a further domain change. Corvi et al. evaluated GAN-era forensic detectors on diffusion imagery and included compression and resizing in their analysis [2]; DIRE instead uses the residual between an image and its diffusion reconstruction as forensic evidence [6].

Recent work has also changed the representation used by the detector. Ojha et al. use fixed features from a large vision-language model to improve transfer beyond a narrow training domain [4]. CLIP-based detectors extend this direction with lightweight training and improved performance on impaired images [7]. These methods address representation and generator transfer. Here, the representation is held fixed so that changes in robustness can be assigned to the training distortions.

B. Post-Processing Robustness

GenImage includes cross-generator tests together with degraded-image conditions such as low resolution, blur, and compression [3]. The full NTIRE 2026 benchmark

Controlled design: only the training augmentation changes

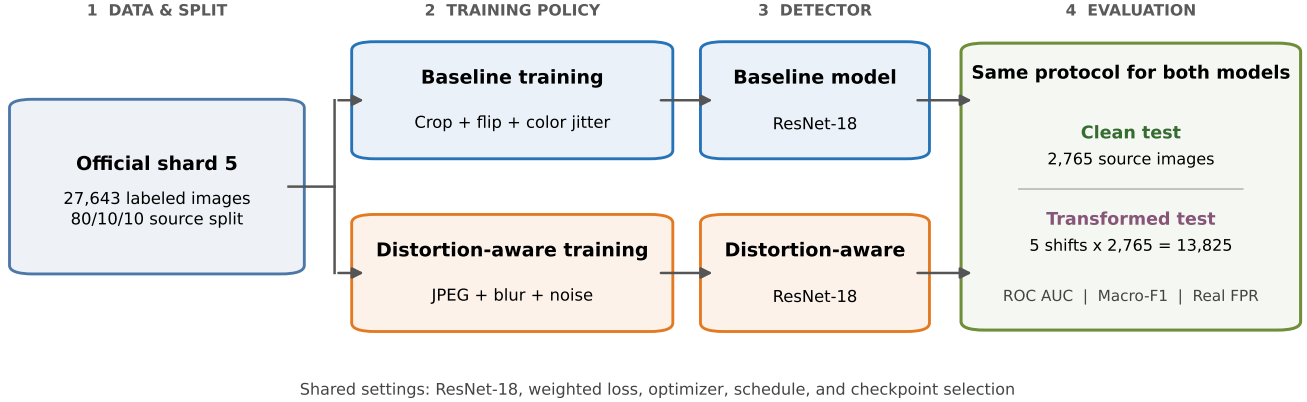


Fig. 1. Controlled experiment. Both detectors share the source-image split, ResNet-18 architecture, weighted loss, optimizer, schedule, and evaluation sets. The baseline uses crop, flip, and color jitter; distortion-aware training additionally applies JPEG compression, Gaussian blur, and Gaussian noise.

broadens this setting to 42 generators and 36 transformations and uses ROC AUC on a mixture of transformed and untransformed images [8]. Competitive systems combine larger training collections, multiple resolutions, strong augmentation, and heterogeneous backbones; HEDGE is one example [9]. The present experiment instead uses one official shard and one compact ResNet-18 to measure the incremental effect of distortion-aware training against a specified baseline.

The severity sweeps follow the corruption-family evaluation used by ImageNet-C [10]. ROC AUC is reported separately from threshold-dependent metrics because ranking can remain strong even when a fixed operating threshold is poorly calibrated under shift [11], [12].

III. Dataset and Evaluation Protocol

A. Source Images and Split

The full NTIRE 2026 benchmark contains 108,750 real images, 185,750 generated images, 42 generators, and 36 transformations [8]. The experiments use official training shard 5, which contains 27,643 unique JPEG source images: 9,947 real and 17,696 AI-generated. A stratified 80/10/10 split with seed 42 assigns 22,113 images to training, 2,765 to validation, and 2,765 to the clean test. Table I lists the class counts. Transformations are generated only after the source split, and every derived copy remains in the split of its source image.

The shard labels specify only whether an image is real or AI-generated. They do not identify the generator. The protocol can enforce source separation and evaluate post-processing shift, but it cannot measure transfer to held-out generators.

TABLE I
Source-image split used in the experiments.

Split	Real	AI	Total
Train	7,957	14,156	22,113
Validation	995	1,770	2,765
Clean test	995	1,770	2,765
Transformed test [†]	4,975	8,850	13,825

[†]Five transformed copies of every clean-test source.

B. Fixed Transformations and Severity Sweeps

The fixed transformed test is built from the 2,765 clean test sources. Five operations are applied independently: JPEG compression at quality 35; Gaussian blur with $\sigma = 1.5$; additive Gaussian noise with standard deviation 0.04 on a $[0, 1]$ scale; downsampling to 50% followed by restoration to the original dimensions; and a center crop that retains 80% of the width and height followed by resizing. The resulting set contains 13,825 images, five transformed versions of every source, with the original binary labels. Figure 6 illustrates the five operations on one real and one AI-generated source.

A separate sweep evaluates five levels for each transformation: JPEG qualities {95, 75, 55, 35, 20}; blur $\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$; noise standard deviations {0.01, 0.02, 0.04, 0.06, 0.08}; downsampling factors {0.90, 0.75, 0.50, 0.35, 0.25}; and retained center-crop fractions {0.95, 0.90, 0.80, 0.70, 0.60}. Inference then resizes the short side to 256 pixels, takes a 224×224 center crop, and applies ImageNet normalization. For the fixed transformed test, non-JPEG outputs are saved once at JPEG quality 95, while the JPEG condition is saved at quality 35. Severity transformations are applied in memory during evaluation. Two training policies, three seeds, five transformation families, and five levels produce

150 severity conditions.

IV. Method

A. ResNet-18 Baseline

Both conditions use an ImageNet-pretrained ResNet-18 [13]. For an input x_i with label $y_i \in \{0, 1\}$, the network produces $p_\theta(y | x_i)$. Its 512-dimensional global-average-pooled feature is passed through dropout with probability 0.20 and a two-output linear layer. All backbone parameters are fine-tuned at 224×224 resolution. Training uses weighted cross entropy,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log p_\theta(y_i | T(x_i)), \quad (1)$$

with $w_{\text{real}} = 1.389531$ and $w_{\text{AI}} = 0.781047$.

The baseline policy applies RandomResizedCrop with scale 0.7–1.0 and aspect ratio 0.8–1.25, horizontal flipping, and ColorJitter with brightness 0.15, contrast 0.15, saturation 0.10, and hue 0.03, followed by ImageNet normalization.

B. Distortion-Aware Training

The distortion-aware policy retains every baseline operation and adds three stochastic transformations. JPEG compression is applied with probability 0.35 at a quality sampled from 35–95. Gaussian blur is applied with probability 0.25 and $\sigma \in [0.1, 2.0]$. Gaussian noise is applied with probability 0.20 and a standard deviation sampled up to 0.04. This is the only difference between the two main training conditions. Table II summarizes the controlled comparison.

Both conditions are optimized for 10 epochs with AdamW [14], batch size 32, learning rate 10^{-4} , weight decay 10^{-4} , one warm-up epoch, cosine decay to 1% of the initial learning rate, and mixed-precision training. The checkpoint with the highest validation ROC AUC is retained. The complete comparison is repeated with seeds 42, 43, and 44.

V. Experiments and Results

A. Metrics and Reproducibility

ROC AUC is the primary metric because it evaluates score ranking over all thresholds and matches the NTIRE challenge metric [8]. Macro-F1 weights the real and AI classes equally. Real-image false-positive rate (FPR) records the proportion of authentic images classified as AI. For each checkpoint, a decision threshold is selected on the validation set by maximizing Youden’s statistic $J = \text{TPR} - \text{FPR}$ and is then held fixed for the clean and transformed tests.

The robustness gap is

$$G = 100 (\text{AUC}_{\text{clean}} - \text{AUC}_{\text{transformed}}), \quad (2)$$

in percentage points. The pooled transformed AUC is computed over all 13,825 transformed instances; it is not

the arithmetic mean of the five transformation-specific AUCs. We report the mean and sample standard deviation over three seeds. For the seed-42 model pair, 2,000 paired bootstrap resamples are drawn at the source-image level [15], keeping the five transformed copies of each source together.

Training and evaluation used PyTorch 2.13.0 with CUDA 13.0 on an NVIDIA RTX 5070 Ti with 16 GB VRAM, an AMD Ryzen 7 9800X3D, and 32 GB system memory. Seed-matched runs use identical manifests, initial pretraining, optimizer settings, schedules, and checkpoint rules.

B. Baseline Degradation and Main Comparison

Table III reports the three-seed comparison. The baseline reaches 97.30% clean AUC and 94.95% pooled transformed AUC, a drop of 2.35 *pp*. Its transformed Macro-F1 is 87.20%, and 12.72% of real transformed images cross the validation-selected threshold.

Distortion-aware training reaches 96.51% clean AUC and 95.90% transformed AUC. Relative to the baseline, clean AUC decreases by 0.79 *pp*, transformed AUC increases by 0.95 *pp*, and the robustness gap falls by 1.74 *pp* to 0.61 *pp*. Transformed Macro-F1 rises by 1.33 *pp*, while real-image FPR falls by 1.44 *pp*.

Figure 2 reports the seed-42 paired bootstrap comparison. The 95% intervals exclude zero for all three contrasts: clean AUC -1.03 *pp* (95% CI $[-1.52, -0.58]$), transformed AUC $+0.91$ *pp* ($[+0.40, +1.40]$), and gap reduction $+1.94$ *pp* ($[+1.70, +2.20]$). Figure 3 reports the seed-matched effects for all three training runs. Each seed records lower clean AUC, higher transformed AUC, and a smaller robustness gap after distortion-aware training.

C. Results by Transformation

The pooled improvement is not uniform across transformations. Figure 4 reports distortion-aware minus baseline effects over three seeds. Blur AUC changes by $+0.04$ *pp*, while blur Macro-F1 rises 1.36 *pp* and its real-image FPR falls 11.66 *pp*. Noise gains 0.75 *pp* AUC and 7.12 *pp* Macro-F1, although its real-image FPR increases by 6.37 *pp*. Downsampling AUC changes by -0.23 *pp* while real-image FPR improves by 6.97 *pp*. Center crop and JPEG lose 0.97 *pp* and 0.65 *pp* AUC, respectively.

Pooled ROC AUC ranks all 13,825 transformed instances jointly and therefore also reflects score ordering across transformation families. The separate scores in Fig. 4 locate the conditions that contribute to, or work against, the pooled result.

D. Severity Response

Figure 5 follows each transformation from mild to severe. The distortion-aware advantage grows most strongly under Gaussian noise. At the highest tested noise level, AUC is 95.08% versus 90.40% for the baseline, a 4.68 *pp* gain. The strongest blur and downsampling conditions

TABLE II
Controlled training configuration. Entries marked “same” are identical to the baseline.

Component	Baseline	Distortion-aware training
Backbone and input	ImageNet ResNet-18, 224×224	Same
Head and objective	Dropout 0.20, linear head, weighted cross entropy	Same
Common augmentation	Crop, flip, color jitter	Same
Added distortions	None	JPEG ($p = 0.35$), blur ($p = 0.25$), noise ($p = 0.20$)
Optimization	AdamW, batch 32, 10 epochs, warm-up + cosine decay	Same
Model selection	Highest validation ROC AUC	Same

TABLE III
Baseline and distortion-aware results on the clean and pooled transformed tests (mean $\pm$ sample standard deviation over seeds 42, 43, and 44, in percent).

Training	Clean AUC $\uparrow$	Trans. AUC $\uparrow$	AUC gap $\downarrow$	Trans. Macro-F1 $\uparrow$	Trans. real FPR $\downarrow$
Baseline	97.30 $\pm$ 0.11	94.95 $\pm$ 0.26	2.35 $\pm$ 0.20	87.20 $\pm$ 0.72	12.72 $\pm$ 0.37
Distortion-aware	96.51 $\pm$ 0.26	95.90 $\pm$ 0.26	0.61 $\pm$ 0.03	88.54 $\pm$ 0.46	11.28 $\pm$ 0.46

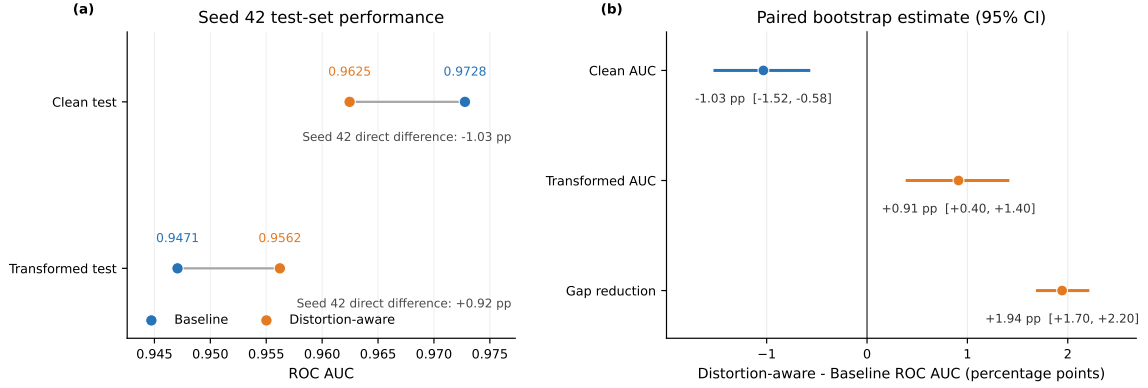


Fig. 2. Seed-42 paired comparison. The left panel compares baseline and distortion-aware AUC on the clean and transformed tests; gray segments show the within-test difference. The right panel reports distortion-aware-minus-baseline effects; bars are 95% source-level paired-bootstrap confidence intervals.

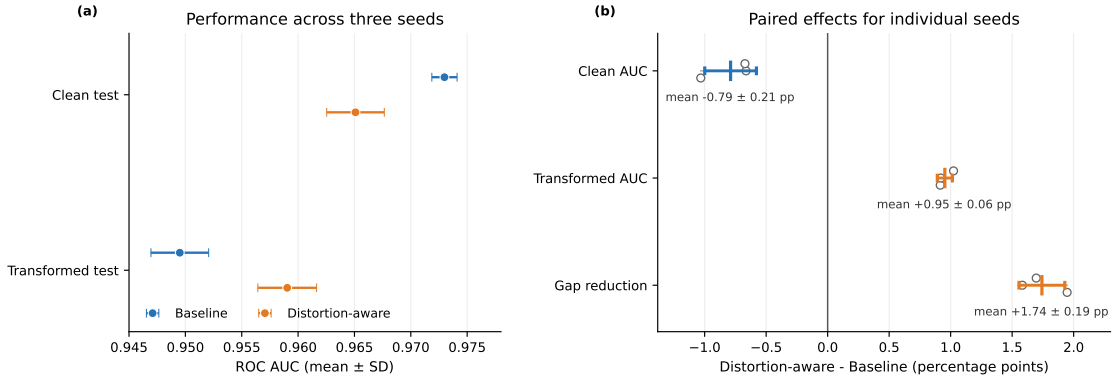


Fig. 3. Three-seed stability. Panel (a) reports absolute clean and transformed AUC as the mean and one sample standard deviation. Panel (b) shows individual seed-matched effects together with their mean and sample standard deviation. Positive transformed-AUC and gap-reduction values favor distortion-aware training.

gain 1.30 *pp* and 0.67 *pp*, respectively. Averaged over the five levels, the effects are +1.56 *pp* for noise, +0.06 *pp* for blur, and -0.21 *pp* for downsampling. JPEG and crop remain negative on average (-0.63 *pp* and -0.84 *pp*).

The advantage is concentrated at high noise and blur severity. Exposure to a transformation during training is not sufficient by itself: the JPEG policy does not improve the JPEG sweep, and center crop remains a

Distortion-aware training effect by evaluation transformation (mean $\pm$ SD, $n = 3$ seeds)

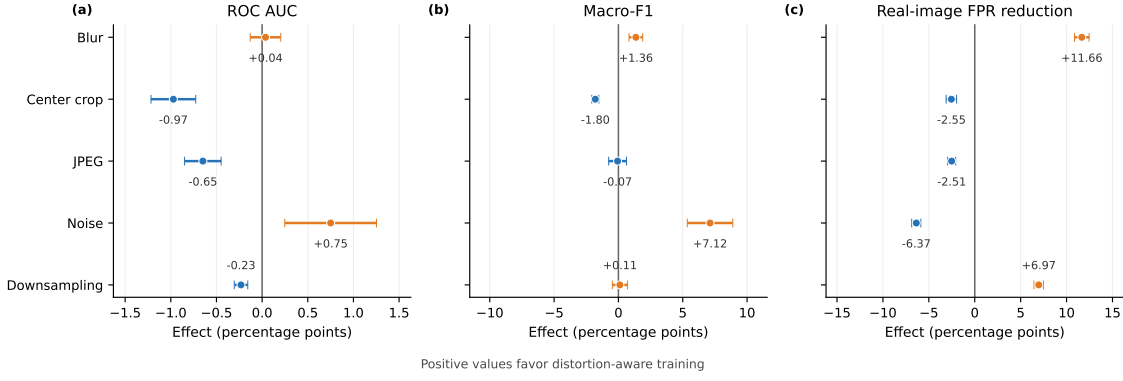


Fig. 4. Change from baseline by transformation, averaged over three seeds. Columns report ROC AUC, Macro-F1, and real-image FPR reduction in percentage points; points are means and bars show one sample standard deviation. Positive values favor distortion-aware training.

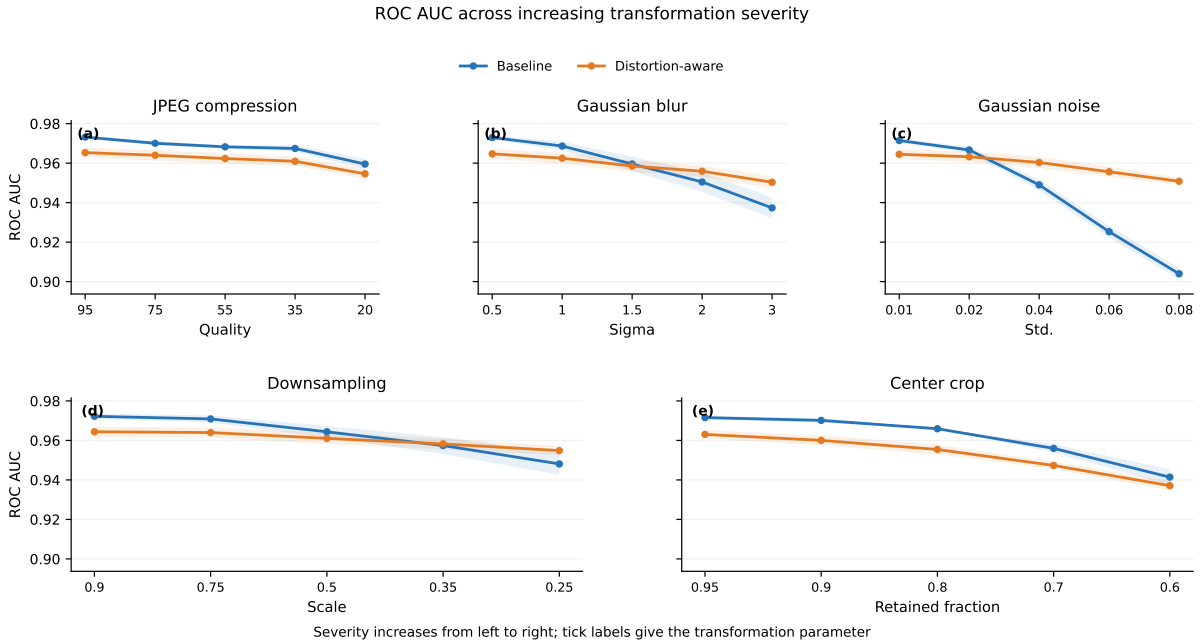


Fig. 5. ROC AUC for the baseline and distortion-aware model over five severity levels. Lines are three-seed means, and bands show one sample standard deviation. Severity increases from left to right.

failure mode. Downsampling was absent from the three added distortions, yet records a small gain at the strongest level, which suggests limited transfer across distortion families.

E. Exploratory Single-Component Comparison

The seed-42 comparison in Table IV replaces the combined policy with JPEG-only, blur-only, or noise-only training. All three single-component policies improve pooled transformed AUC relative to the baseline. Blur-only records the highest transformed AUC (95.93%) and the lowest real-image FPR (9.71%) on this seed. Noise-only reaches 95.83% transformed AUC but also the highest

real-image FPR (15.98%). The combined policy has the smallest clean-to-transformed gap (0.62 *pp*). Because this comparison uses one seed, its role is descriptive.

VI. Discussion

The baseline quantifies the effect of routine post-processing on score ranking. Distortion-aware training recovers part of the transformed-set performance, but the exchange is asymmetric across input conditions. A mostly pristine stream favors the baseline’s clean ranking, while a stream containing strong noise or blur places more value on distortion-aware training.

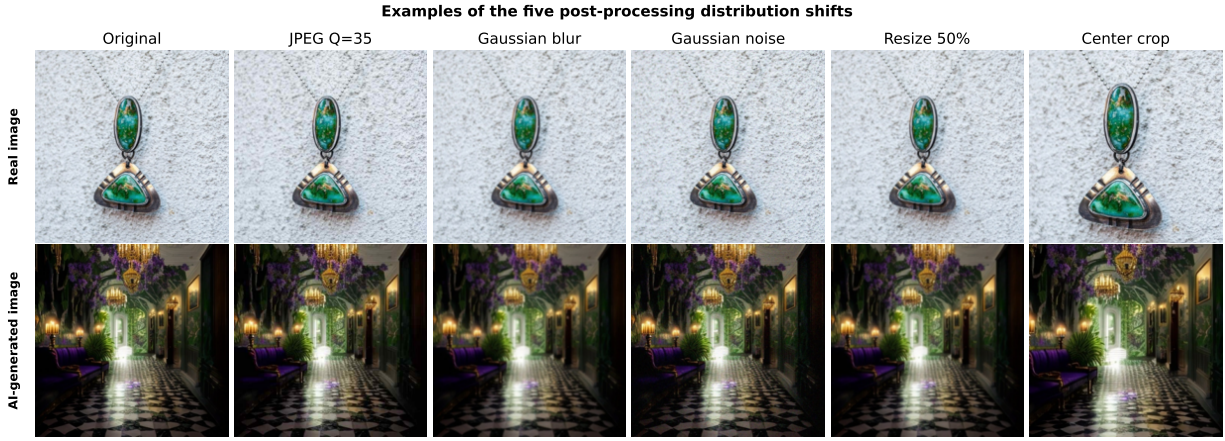


Fig. 6. Two held-out sources before and after the five fixed post-processing operations. Each row preserves source identity and class label.

TABLE IV
Single-component training comparison for seed 42 (percent).

Training	Clean	Trans.	Gap	Real FPR
Baseline	97.28	94.71	2.57	12.50
JPEG only	97.05	95.24	1.82	11.02
Blur only	97.18	95.93	1.26	9.71
Noise only	97.14	95.83	1.31	15.98
Combined	96.25	95.62	0.62	10.87

Ranking and threshold metrics describe different aspects of the model. Under noise, distortion-aware training improves AUC and Macro-F1 while increasing the proportion of real images that cross the fixed threshold. Under blur, AUC changes little while real-image FPR falls markedly. ROC AUC measures ordering over all thresholds, whereas FPR is tied to one operating point. Deployment would require a threshold selected for the expected class balance and the relative costs of false accusations and missed synthetic images.

Grad-CAM [16] provides a qualitative diagnostic in Fig. 7. In panel (a), distortion-aware training corrects a blurred real image that the baseline assigns to the AI class. In panel (b), it introduces the opposite error on a cropped AI image. In these examples, the attribution maps differ between checkpoints, consistent with a change in the spatial evidence used for prediction. These maps are case-level diagnostics; the quantitative comparisons above provide the evidence for performance.

VII. Limitations

The study covers 27,643 source images from one of the six official training shards. Its content distribution may differ from the remaining shards and from images encountered after deployment. Generator identities are absent from the labels used here. The results measure post-processing robustness for a source-separated binary split, while a temporal benchmark such as AI-GenBench is designed to study future generators [17].

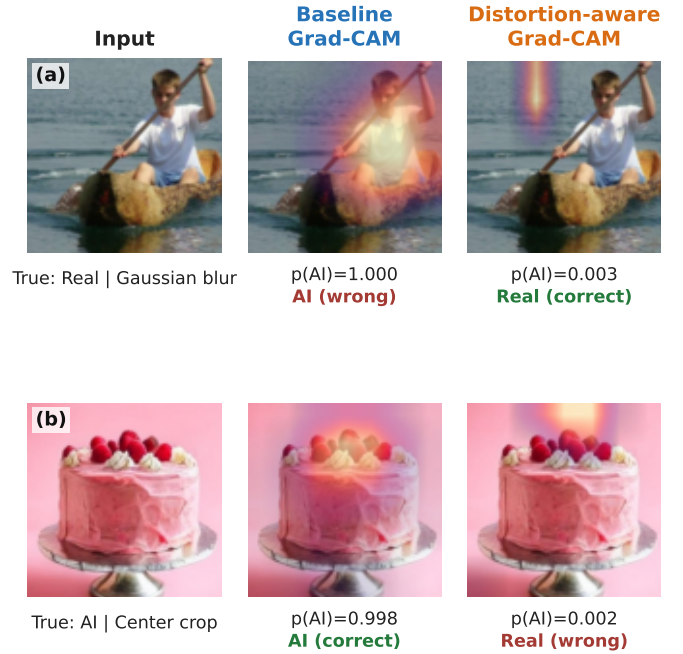


Fig. 7. Seed-42 Grad-CAM maps for two predictions changed by distortion-aware training: (a) a blurred real image corrected by the model and (b) a cropped AI image newly misclassified as real. Warmer regions indicate stronger spatial evidence.

The comparison uses one ResNet-18 at 224×224 . Higher-resolution inputs, CLIP features, or heterogeneous ensembles may respond differently to the same augmentation policy [7], [9]. Each transformed test image also contains one synthetic operation, whereas platform pipelines may apply several operations in sequence and introduce codec, content, and format biases. Challenging-set studies similarly show that benchmark accuracy does not determine open-world reliability [5].

The main baseline-versus-combined comparison uses three seeds, but the component analysis and Grad-CAM examples use seed 42 only. The evidence in this report

is therefore strongest for the three-seed AUC, Macro-F1, FPR, and severity comparisons on NTIRE 2026 shard 5.

VIII. Conclusion

The ResNet-18 baseline achieved 97.30% clean AUC and 94.95% pooled transformed AUC on NTIRE 2026 shard 5. Adding JPEG compression, Gaussian blur, and Gaussian noise to the same training pipeline changed these values to 96.51% and 95.90%. Across three seeds, distortion-aware training exchanged 0.79 *pp* of clean AUC for a 0.95 *pp* transformed-AUC gain and a 1.74 *pp* reduction in the clean-to-transformed gap. The improvement was largest under severe noise and blur; JPEG compression and center crop did not improve consistently. For this model and shard, distortion-aware training is most useful when the expected input stream contains the severe noise and blur conditions for which gains were observed. Evaluation across generator-aware splits, additional backbones, and composed platform processing is the next step.

References

- [1] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, “CNN-generated images are surprisingly easy to spot... for now,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 8695–8704.
- [2] R. Corvi, D. Cozzolino, G. Zingarini, G. Poggi, K. Nagano, and L. Verdoliva, “On the detection of synthetic images generated by diffusion models,” in IEEE International Conference on Acoustics, Speech and Signal Processing, 2023, pp. 1–5.
- [3] M. Zhu, H. Chen, Q. Yan, X. Huang, G. Lin, W. Li, Z. Tu, H. Hu, J. Hu, and Y. Wang, “GenImage: A million-scale benchmark for detecting AI-generated image,” in Advances in Neural Information Processing Systems, vol. 36, 2023, pp. 77 771–77 782.
- [4] U. Ojha, Y. Li, and Y. J. Lee, “Towards universal fake image detectors that generalize across generative models,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 24 480–24 489.
- [5] S. Yan, O. Li, J. Cai, Y. Hao, X. Jiang, Y. Hu, and W. Xie, “A sanity check for AI-generated image detection,” in The Thirteenth International Conference on Learning Representations, 2025. [Online]. Available: <https://openreview.net/forum?id=ODRHZrkOQM>
- [6] Z. Wang, J. Bao, W. Zhou, W. Wang, H. Hu, H. Chen, and H. Li, “DIRE for diffusion-generated image detection,” in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 22 445–22 455.
- [7] D. Cozzolino, G. Poggi, R. Corvi, M. Nießner, and L. Verdoliva, “Raising the bar of AI-generated image detection with CLIP,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2024, pp. 4356–4366.
- [8] A. Gushchin, K. Abud, E. Shumitskaya, A. Filippov, G. Bychkov, S. Lavrushkin et al., “NTIRE 2026 challenge on robust AI-generated image detection in the wild,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2026, pp. 1895–1913.
- [9] F. Wu, D. Lu, M. Yao, X. Xu, and F. Guo, “HEDGE: Heterogeneous ensemble for detection of AI-generated images in the wild,” arXiv preprint arXiv:2604.03555, 2026.
- [10] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in International Conference on Learning Representations, 2019. [Online]. Available: <https://openreview.net/forum?id=HJz6tiCqYm>
- [11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in Proceedings of the 34th International Conference on Machine Learning, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330.
- [12] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, and J. Snoek, “Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift,” in Advances in Neural Information Processing Systems, vol. 32, 2019.
- [13] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [14] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in International Conference on Learning Representations, 2019. [Online]. Available: <https://openreview.net/forum?id=Bkg6RiCqY7>
- [15] B. Efron, “Bootstrap methods: Another look at the jackknife,” The Annals of Statistics, vol. 7, no. 1, pp. 1–26, 1979.
- [16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 618–626.
- [17] L. Pellegrini, D. Cozzolino, S. Pandolfini, D. Maltoni, M. Ferrara, L. Verdoliva, M. Prati, and M. Ramilli, “AI-GenBench: A new ongoing benchmark for AI-generated image detection,” in 2025 International Joint Conference on Neural Networks, 2025, pp. 1–9.